

Confining a Language Model to a Prescribed Role: An Adversarial Reliability Test for Instruction Chatbots in Economics Experiments

Can Çelebi · Preliminary, June 1, 2026

Language models are increasingly proposed to replace the human assistants who answer participants' questions during economics experiments, where the assistant must explain a task's rules while withholding anything that could influence the decisions under study; the reliability question is therefore not whether a model is helpful but how dependably it can be confined to a prescribed role across an adversarial, multi-turn exchange. We study this in Oprea's (2024) experiments on decision-making under risk, whose interpretation turns on distinguishing a participant's *comprehension* of a lottery from the *complexity* of evaluating it: an assistant there must reduce comprehension costs—restating rules, defining terms—without reducing complexity costs, and so may not give advice, compute or hint at expected values or dominance, simplify the task, signal the experimenter's intent, or reveal other treatments. We implement a chatbot bound by these constraints and ask how reliably the constraint holds as a function of system design. Without human participants, we generate adversarial synthetic subjects—seven participant types, each engineered to press a distinct failure vector—and have each conduct fifteen fifteen-turn dialogues under two experimental contexts, the second materially harder; every response is routed into one of six communicative categories and scored by an independent model judge for violation of the no-advice constraint, factual inconsistency, and incoherence. Across this procedure we compare five architectures, from a single constrained call to multi-agent configurations adding classification, low-confidence gating, and post-hoc correction, and, holding the best fixed, vary the base model's capability and reasoning effort—more than twenty-seven thousand scored responses. Architectural elaboration does not improve reliability: the single call is undominated on adherence, cost, and latency, and every added agent matches or worsens the violation rate by restating and thereby re-exposing the prohibited request. Holding architecture fixed, only greater base-model capability improves adherence, cutting violations by roughly forty-six and sixty-eight percent across the two contexts at negligible cost, whereas additional reasoning effort yields nothing and an explicit correction stage is harmful; because small per-response violation probabilities compound over a dialogue, we assess reliability per conversation rather than per turn. These findings are preliminary—a confirmatory condition fully separating the evaluated and evaluating models is still in progress—but already indicate that, for a chatbot held to a narrowly prescribed role, the capability of the base model rather than the elaborateness of the surrounding workflow is the binding determinant of reliability. Full methods and interactive results: can-celebi.github.io/chatbot-for-experiments.